

Excel survival guide for analysis

Everything you need for a descriptive analysis, on one sheet.

Rule zero

WATCH OUT

Work on a copy. Always. Duplicate your data sheet before you touch anything, and never type over raw data. Excel has no audit trail — once you overwrite a cell, the original is gone forever.

Descriptive statistics

What you want	Formula
Count of numeric values (your denominator!)	=COUNT(D2:D181)
Count of non-empty cells including text	=COUNTA(D2:D181)
Count of blanks	=COUNTBLANK(D2:D181)
Mean	=AVERAGE(D2:D181)
Median	=MEDIAN(D2:D181)
Standard deviation (sample)	=STDEV.S(D2:D181)
Minimum / Maximum	=MIN(D2:D181) =MAX(D2:D181)
First and third quartile	=QUARTILE.INC(D2:D181,1) =QUARTILE.INC(D2:D181,3)
Count matching one condition	=COUNTIF(\$Q\$2:\$Q\$181,"Cured")
Count matching two conditions	=COUNTIFS(\$E\$2:\$E\$181,"Male",\$Q\$2:\$Q\$181,"Died")
A percentage, safely	=COUNTIF(...)/COUNT(...) then format as %
Mean of one group only	=AVERAGEIF(\$E\$2:\$E\$181,"Female",\$D\$2:\$D\$181)

TIP

Use \$ to lock your ranges (\$D\$2:\$D\$181) before you copy a formula down a column. Forgetting this is the most common reason an Excel table full of percentages is quietly wrong.

PivotTable in six steps — the fastest route to a results table

1. Click any cell inside your data.
2. **Insert > PivotTable > OK** (it goes on a new sheet).
3. Drag **outcome** into the **Rows** box.
4. Drag **record_id** into the **Values** box — it will show *Count of record_id*.
5. To get percentages: right-click any number > **Show Values As > % of Grand Total**.
6. To cross-tabulate, drag **sex** into **Columns**. You now have a two-way table.

The pivot table trap

A PivotTable **silently drops blank cells** from its row categories. If your outcome column has 8 blanks, your pivot shows

172, not 180 — and gives you no warning. Always check the grand total against the number of records you expect.

Making a chart that is not embarrassing

1. Select your **summary table** (from the pivot), not the raw data.
2. **Insert > Charts >** Clustered Column (bar), Line, or Scatter.
3. Delete immediately: the chart border, the gridlines, the legend if there is only one series.
4. Click the chart title and type a real title: what, who, where, when, and n.
5. **Chart Design > Add Chart Element > Axis Titles** — add units to both axes.
6. Right-click the y-axis > **Format Axis** > check that **Minimum is 0**.
7. Sort your bars largest to smallest (sort the source table) unless there is a natural order.

Histogram (for a continuous variable)

Excel 2016 and later: select your column, then **Insert > Charts > Histogram**. Right-click the horizontal axis > **Format Axis** > set **Bin width** to something sensible (5 years for age, 5 km for distance). Excel's automatic bins are usually poor.

Liafuan importante • Key words in Tetum

Formula	Fórmula
Cell	Sélula
Sheet	Fólia
Sort	Ordena
Copy	Kopia
Save	Rai